

Aljon A. Novelo

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OBJECTIVE

- To participate as a software engineer in a work environment focused on promoting digital transformation in businesses.

WORK EXPERIENCE

Web Developer Intern

i-BRIDGE Systems Philippines Inc.

July - August 2025

- Contributed to **Agile sprint management and development workflows** using GitLab, Redmine, and Microsoft Office for collaboration, task tracking, and project documentation
- Designed and implemented **additional user interface components** for an internal tool using the MERN stack
- Developed and integrated **additional database fields** to support the new components
- Conducted **preliminary unit and integration testing** using Mocha and Chai to identify and resolve issues before deployment
- Deployed the new features to **Google Cloud Platform**

PROJECTS

Learning Platform for LaTeX Writers | Web

- Conducted **contextual interviews with 20 students** to identify requirements for a LaTeX-focused learning platform
- Developed the platform using the **MERN stack**
- Implemented a **recursive-descent parser** to process LaTeX input
- Achieved a **System Usability Scale score of 83.25** and performed **thematic analysis** of interview responses to identify areas for improvement

Color-Based Image Segmentation | Research

- Implemented a **K-means clustering algorithm in C++** to segment images based on dominant color clusters
- Processed **1,000+ sequential JPEG frames**
- Demonstrated the project through a **video presentation** showcasing unsupervised clustering for image segmentation

Context-Free Grammar Simulator | Research

- Developed a **Python-based command-line simulator** that performs **LL parsing**
- Designed the simulator to support **unambiguous grammars**
- Implemented **file-based input parsing** for grammar definitions

SKILLS

Languages: JavaScript, Python, Java, C, Dart, R, Assembly, C++

Databases: MongoDB, PostgreSQL, MariaDB, Firebase

Tools: GitLab, Google Cloud Platform, GitHub, Postman, Figma

EDUCATIONAL BACKGROUND

Bachelor of Science in Computer Science

University of the Philippines - Los Baños

September 2021 - July 2026

ORGANIZATIONAL EXPERIENCE

Volunteer

UPLB Ugnayan ng Pahinungód

March 2023 - Present

- Co-facilitated training sessions for **140+ new volunteers**
- Completed **training on mental health and psychosocial support** for elementary students
- Facilitated **peer-to-peer reading sessions** for Grade 4-6 students
- Facilitated team-building activities for **50+ students**
- Assisted in administering college entrance exam simulations for **Grade 11 students**

Technical Director

UPLB Street Jazz Dance Company

October 2023 - May 2024

- Led a **technical committee** responsible for stage setup, audio, and lighting
- Coordinated with **partner organization** to align production plans

Executive Member

Kapatirang Pitong Lawa sa UP Los Baños

January - April 2023

- Led **socio-cultural activities** of the organizations
- Managed **logistical and technical support** for on-ground community activities
- Coordinated with **organization alumni and partner organizations** for activities